Angles in Triangles and Quadrilaterals

Name _____ Date _____ No calculator Version 1

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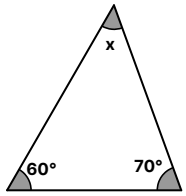
KEY WORDS interior angle · triangle *(tapatoru)* · quadrilateral *(tapawhā)* · angle sum · right angle *(koki hāngai)*

A Angles in a triangle ●○○

Remember: The three angles inside any triangle add to 180°.

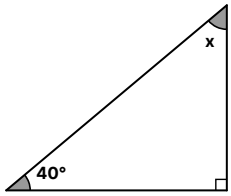
1. Find the size of the angle marked x.

a)



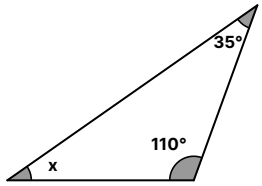
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b)



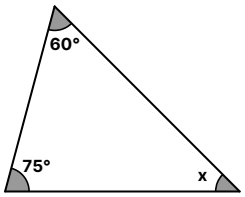
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c)



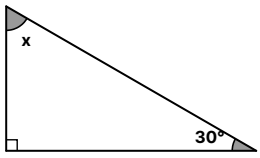
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d)



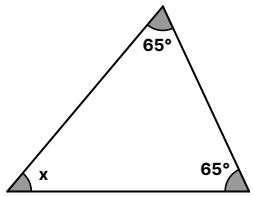
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e)



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f)



_____

2. Answer each question.

a) Two angles of a triangle are 35° and 80°. Find the third angle. _____

b) Two angles of a triangle are 90° and 47°. Find the third angle. _____

c) Two angles of a triangle are 118° and 26°. Find the third angle. _____

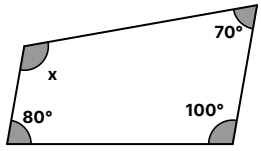
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B Angles in a quadrilateral ●●○

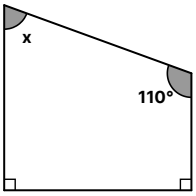
Remember: The four angles inside any quadrilateral add to 360°.

3. Find the size of the angle marked x.

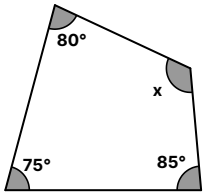
a)



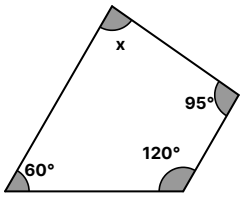
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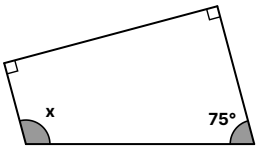
c)



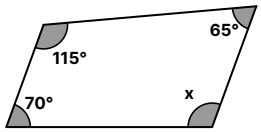
d)



e)



f)



4. Answer each question.

a) Three angles of a quadrilateral are 90°, 85° and 110°. Find the fourth angle. _____

b) Three angles of a quadrilateral are 72°, 108° and 64°. Find the fourth angle. _____

c) Three angles of a quadrilateral are 90°, 90° and 90°. Find the fourth angle. _____

C Could it be drawn? ●●●

Remember: Add the angles. A triangle needs a total of exactly 180°, and a quadrilateral needs exactly 360°.

5. Could a triangle have these three angles? Show your working.

a) 60°, 60°, 60°

b) 90°, 50°, 50°

c) 100°, 45°, 35°

d) 70°, 70°, 30°

6. Could a quadrilateral have these four angles? Show your working.

a) 90°, 90°, 90°, 90°

b) 100°, 80°, 110°, 60°

c) 120°, 60°, 120°, 60°

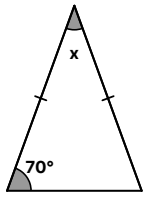
d) 95°, 85°, 100°, 80°

★ Extension

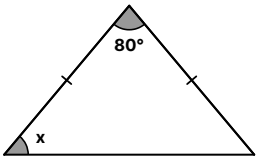
7. One angle of a right-angled triangle is 35°. Find the size of its third angle.

8. The small lines mark the equal sides. Find the size of the angle marked x. Give a reason.

a)



b)



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Angles in Triangles and Quadrilaterals

Name _____ Date _____ No calculator Version 2

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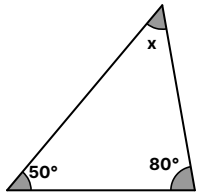
KEY WORDS interior angle · triangle *(tapatoru)* · quadrilateral *(tapawhā)* · angle sum · right angle *(koki hāngai)*

A Angles in a triangle ●○○

Remember: The three angles inside any triangle add to 180°.

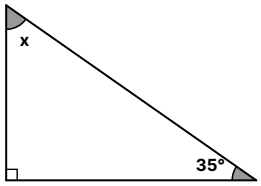
1. Find the size of the angle marked x.

a)



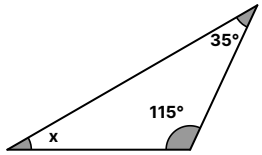
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b)



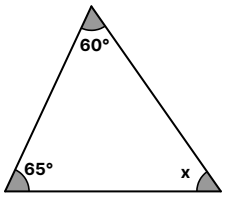
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c)



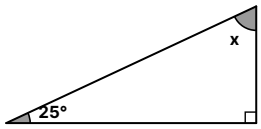
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d)



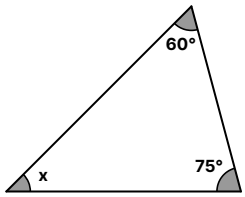
_____

e)



_____

f)



_____

2. Answer each question.

a) Two angles of a triangle are 45° and 70°. Find the third angle. _____

b) Two angles of a triangle are 90° and 53°. Find the third angle. _____

c) Two angles of a triangle are 124° and 19°. Find the third angle. _____

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B Angles in a quadrilateral ●●○

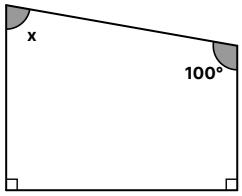
Remember: The four angles inside any quadrilateral add to 360°.

3. Find the size of the angle marked x.

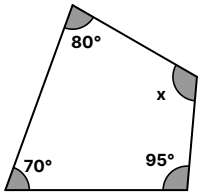
a)



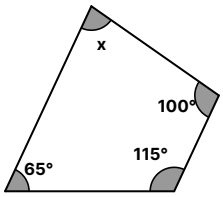
b)



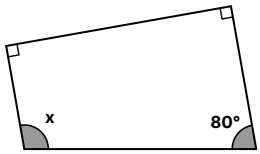
c)



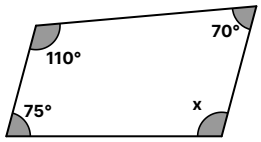
d)



e)



f)



4. Answer each question.

a) Three angles of a quadrilateral are 90°, 95° and 105°. Find the fourth angle. _____

b) Three angles of a quadrilateral are 68°, 112° and 74°. Find the fourth angle. _____

c) Three angles of a quadrilateral are 90°, 90° and 90°. Find the fourth angle. _____

C Could it be drawn? ●●●

Remember: Add the angles. A triangle needs a total of exactly 180°, and a quadrilateral needs exactly 360°.

5. Could a triangle have these three angles? Show your working.

a) 45°, 45°, 90°

b) 80°, 60°, 50°

c) 110°, 40°, 30°

d) 75°, 75°, 40°

6. Could a quadrilateral have these four angles? Show your working.

a) 90°, 90°, 100°, 80°

b) 110°, 70°, 120°, 50°

c) 130°, 50°, 130°, 50°

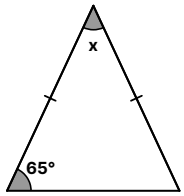
d) 95°, 95°, 80°, 80°

★ Extension

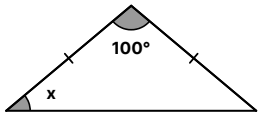
7. One angle of a right-angled triangle is 28°. Find the size of its third angle.

8. The small lines mark the equal sides. Find the size of the angle marked x. Give a reason.

a)



b)



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